

23 Exponential Distributions

Exponential distributions are often used to model the *waiting times* between events in a random process that occurs continuously over time.

Example 23.1. [NASA tracks data on fireball events](#), exceptionally bright meteors that are spectacular enough to be seen over a very wide area¹. Let X be the time, measured in months (assume 30 days), between any two fireballs and suppose X has pdf

$$f_X(x) = 2.5e^{-2.5x}, \quad x \geq 0$$

We say that X has an Exponential distribution with rate 2.5 per month.

1. Sketch the pdf of X and verify that f_X is a valid pdf.
2. Without doing any integration, approximate the probability that the time until the next fireball *rounded to the nearest day* is 6 days.
3. Compute the probability that the time until the next fireball is less than 6 days.
4. Compute and interpret $P(X > 2)$.

¹Thanks for former STAT 305 student Martin Hsu for this [example and data](#).

5. Find the cdf of X .
6. Find the median time between fireballs.
7. Suggest a shortcut formula for $E(X)$. Then compute and interpret $E(X)$. How does the mean compare to the median? Why?
8. Compute and interpret $P(X \leq E(X))$.
9. Compute $E(X^2)$.
10. Find $\text{Var}(X)$ and $\text{SD}(X)$.
11. Suppose Y is the time between fireballs measured in *days*. How does all of the above change?

- A continuous random variable X has an **Exponential distribution** with *rate* parameter $\lambda > 0$ if its pdf is

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & \text{otherwise} \end{cases}$$

- If X has an Exponential(λ) distribution then

$$\text{cdf: } F_X(x) = 1 - e^{-\lambda x}, \quad x \geq 0$$

$$P(X > x) = e^{-\lambda x}, \quad x \geq 0$$

$$\text{quantile: } Q(p) = -(1/\lambda) \ln(1 - p), \quad 0 < p < 1$$

$$E(X) = \frac{1}{\lambda}$$

$$\text{Var}(X) = \frac{1}{\lambda^2}$$

$$\text{SD}(X) = \frac{1}{\lambda}$$

- Percentiles for any Exponential distribution follow a particular pattern in terms of multiples of the mean:
 - For any $m > 0$, $P(X \leq mE(X)) = 1 - e^{-m}$.
 - The p th percentile is $-\ln(1 - p)$ times $E(X)$.
 - See Table [23.1](#).
- Exponential distributions are often used to model the *waiting time* in a random process until some event occurs.
 - λ is the average *rate* at which events occur over time (e.g., 2 per hour)
 - $1/\lambda$ is the mean time between events (e.g., 1/2 hour)
- If X has an Exponential(λ) distribution and $a > 0$ is a constant, then aX has an Exponential(λ/a) distribution.
 - If X is measured in hours with rate $\lambda = 2$ per hour and mean 1/2 hour
 - Then $60X$ is measured in minutes with rate $2/60$ per minute and mean $60(1/2) = 30$ minutes.
- If X has an Exponential(1) distribution and $\lambda > 0$ is a constant then X/λ has an Exponential(λ) distribution.
- If U has a Uniform(0, 1) distribution then $-\ln(1 - U)$ has Exponential(1) distribution, and $-(1/\lambda) \ln(1 - U)$ has an Exponential(λ) distribution.

Table 23.1: Percentiles for an Exponential distribution in terms of multiples of the mean

Table 23.2: Probabilty pattern for Exponential distributions

Percentile	Relative to mean
10%	0.11 times the mean
25%	0.29 times the mean
39.3%	0.50 times the mean
50%	0.69 times the mean
63.2%	1.00 times the mean
75%	1.39 times the mean
77.7%	1.50 times the mean
86.5%	2.00 times the mean
90%	2.30 times the mean
91.8%	2.50 times the mean
95%	3.00 times the mean
97%	3.50 times the mean
98.2%	4.00 times the mean
98.9%	4.50 times the mean
99.3%	5.00 times the mean

Example 23.2. Continuing Example 23.1. Let X be the waiting time (months) until the next fireball and assume X has an Exponential(2.5) distribution.

1. Find the conditional probability that the waiting time from now until the next fireball is greater than 3 months given that no fireballs occur in the next month. Be sure to write a valid probability statement involving X before computing.
2. Compare to $P(X > 2)$. What do you notice?

- **Memoryless property.** If W has an Exponential(λ) distribution then for any $w, h > 0$

$$P(W > w + h \mid W > w) = P(W > h)$$

- Given that we have already waited w units of time, the conditional probability that we wait at least an additional h units of time is the same as the unconditional probability that we wait at least h units of time.
- In this sense, the future waiting time does not “remember” how long we have already waited.
- A continuous random variable W has the memoryless property if and only if W has an Exponential distribution. That is, Exponential distributions are the *only* continuous² distributions with the memoryless property.
- If W has an $\text{Exponential}(\lambda)$ distribution then the conditional distribution of the *excess* waiting time, $W - w$, given $\{W > w\}$ is $\text{Exponential}(\lambda)$.

Example 23.3. Xiomara and Rogelio each leave work at noon from different locations to meet the other for lunch. The amount of time, X , that it takes Xiomara to arrive is a random variable with an Exponential distribution with mean 10 minutes. The amount of time, Y , that it takes Rogelio to arrive is a random variable with an Exponential distribution with mean 20 minutes. Assume that X and Y are independent. Let W be the time, in minutes after noon, at which the first person arrives.

1. What is the relationship between W and X, Y ?
2. Compute and interpret $P(W > 40)$.
3. Find $P(W > w)$ and identify by name the distribution of W .
4. Compute and interpret $E(W)$. Is it equal to $\min(E(X), E(Y))$?

²Geometric distributions are the only discrete distributions with the discrete analog of the memoryless property.

5. Is $P(Y > X)$ greater than or less than 0.5? Make an educated guess for $P(Y > X)$. Then run a simulation to approximate the probability.

6. Use simulation to approximate the conditional distribution of W given $\{Y > X\}$ and the conditional distribution of W given $\{Y < X\}$. What do you notice?

- **Exponential race** (a.k.a., competing risks.) Let $W_1, W_2, \dots, W_n$ be independent random variables. Suppose W_i has an Exponential distribution with rate parameter λ_i . Let $W = \min(W_1, \dots, W_n)$ and let I be the discrete random variable which takes value i when $W = W_i$, for $i = 1, \dots, n$. Then
 - W has an Exponential distribution with rate $\lambda = \lambda_1 + \dots + \lambda_n$
 - $P(I = i) = P(W = W_i) = \frac{\lambda_i}{\lambda_1 + \dots + \lambda_n}, i = 1, \dots, n$
 - W and I are independent
- Imagine there are n contestants in a race, labeled $1, \dots, n$, racing independently, and W_i is the time it takes for the i th contestant to finish the race. Then $W = \min(W_1, \dots, W_n)$ is the winning time of the race, and W has an Exponential distribution with rate parameter equal to sum of the individual contestant rate parameters.
- The discrete random variable I is the label of which contestant is the winner. The probability that a particular contestant is the winner is the contestant's rate divided by the total rate. That is, the probability that contestant i is the winner is proportional to the contestant's rate λ_i .
- Furthermore, W and I are independent. Information about the winning time does not influence the probability that a particular contestant won. Information about which contestant won does not influence the distribution of the winning time.

Example 23.4. Database queries to the Cal Poly data warehouse occur randomly throughout the day. During regular business hours, queries arrive at rate 0.8 per second on average, so that the average number of queries that arrive during any t second time interval is $0.8t$. Suppose that the *number* of queries that arrive during any t second time interval follows a marginal Poisson distribution with mean $0.8t$.

We are interested in the distribution of T , the *time* (seconds) until the next query arrives.

1. Interpret the event $\{T > 2\}$. How can you express this as an equivalent event involving the number of queries?

2. Compute $P(T > 2)$.

3. Compute $P(T > t)$ as a function of $t > 0$.

4. Identify by name the distribution of T ; be sure to specify the values of any relevant parameters.

5. Compute and interpret $E(T)$.

- Events occur continuously over time according to a **Poisson process** with *rate* $\lambda > 0$ (per unit time) if
 - The number of events that occur in any time interval of length t has a $\text{Poisson}(\lambda t)$ distribution.
 - * In particular, the distribution does not depend on the starting time or ending time of the interval, just its length.
 - The numbers of events that occur in disjoint intervals of time are independent.
- A random process is a Poisson process with rate λ if and only if

- it is a counting process, that is, it is a process which counts the number of occurrences of some event over time.
- the *interarrival times* (times between events) $W_0, W_1, \dots$ are independent $\text{Exponential}(\lambda)$ random variables
- A random process is a Poisson process with rate λ if and only if
 - It is a counting process
 - The numbers of events that occur in disjoint intervals of time are independent
 - For small h ,

$$P(0 \text{ events in time interval of length } h) \approx 1 - \lambda h$$

$$P(1 \text{ events in time interval of length } h) \approx \lambda h$$

$$P(\text{at least 2 events in time interval of length } h) \approx 0$$

- * Roughly, in short intervals of time, at most one event will occur, and the probability that an event does occur is proportional to the length of the time interval.
- * That is, if time is chopped up into many short intervals of equal length h , a Poisson Process behaves like a Bernoulli process: each short interval is a trial, an event either occurs (success) or not, the trials (intervals) are independent, the probability of success is the same for each trial (λh), and the process counts the number of successes

Example 23.5. Suppose that elapsed times (hours) between successive fireballs are independent, each having an $\text{Exponential}(2.5)$ distribution. Let T be the time elapsed from now until the third fireball occurs.

1. Compute $E(T)$.
2. Compute $SD(T)$.
3. Does T have an Exponential distribution? Explain.

4. Use simulation to approximate the distribution of T .

- A continuous random variable X has a **Gamma distribution** with *shape* parameter α , a positive integer³, and *rate* parameter⁴ $\lambda > 0$ if its pdf is

$$\begin{aligned} f_X(x) &= \frac{\lambda^\alpha}{(\alpha-1)!} x^{\alpha-1} e^{-\lambda x}, & x \geq 0, \\ &\propto x^{\alpha-1} e^{-\lambda x}, & x \geq 0 \end{aligned}$$

If X has a $\text{Gamma}(\alpha, \lambda)$ distribution then

$$\begin{aligned} E(X) &= \frac{\alpha}{\lambda} \\ \text{Var}(X) &= \frac{\alpha}{\lambda^2} \\ \text{SD}(X) &= \frac{\sqrt{\alpha}}{\lambda} \end{aligned}$$

- If $W_1, \dots, W_n$ are independent and each W_i has an $\text{Exponential}(\lambda)$ distribution then $(W_1 + \dots + W_n)$ has a $\text{Gamma}(n, \lambda)$ distribution.
- Each W_i represents the waiting time *between* two occurrences of some event, so $W_1 + \dots + W_n$ represents the total waiting time until a total of n occurrences.
- Exponential distributions are continuous analogs of Geometric distributions, and Gamma distributions are continuous analogs of Negative Binomial distributions.
- For a positive integer d , the $\text{Gamma}(d/2, 1/2)$ distribution is also known as the **chi-square distribution with d degrees of freedom**.

³There is a more general expression of the pdf which replaces $(\alpha-1)!$ with the *Gamma function* $\Gamma(\alpha) = \int_0^\infty u^{\alpha-1} e^{-u} du$, that can be used to define a Gamma pdf for any $\alpha > 0$. When α is a positive integer, $\Gamma(\alpha) = (\alpha-1)!$.

⁴Like Exponential distributions, Gamma distributions are sometimes parametrized directly by their mean $1/\lambda$, instead of the rate parameter λ . The mean $1/\lambda$ is called the scale parameter.